Tableau Data Visualization Assignment Report

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Abstract

This report presents the design and implementation of a multi-source Business Intelligence system using Tableau Desktop. An e-commerce dataset comprising multiple related tables — including sales transactions, customer profiles, product catalogs, geographic data, time dimensions, and marketing campaigns — was organized into a star schema and subjected to rigorous data engineering (deduplication, SCD Type 2 implementation, surrogate key assignment). Advanced analytical techniques were applied including Level of Detail (LOD) expressions, cohort retention analysis, exponential smoothing forecasts, K-Means clustering, RFM segmentation, and Z-score anomaly detection. Three interactive dashboards were built: an Executive Control Panel for high-level KPIs, a Customer Intelligence Lab for behavioral segmentation, and an Operational Risk Dashboard for identifying profit leakage and discount abuse. Performance optimization was achieved through extract connections, context filters, and efficient data modeling. The report also addresses key theoretical concepts including Tableau architecture, the Hyper engine, data blending vs. joining, and a comparative analysis of Tableau, Power BI, and Qlik Sense.

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Introduction

Objective

This project implements a multi-source, enterprise-level Business Intelligence system using Tableau Desktop, demonstrating real-world data modeling, advanced analytics, predictive forecasting, cohort analysis, K-Means clustering, and performance optimization.

Tools Used

- **Tableau Desktop** — Primary BI and visualization tool
- **Python** (pandas, numpy) — Data preprocessing and engineering
- **Microsoft Excel** — Data storage format

Datasets

E-commerce multi-table dataset sourced and structured for BI analysis:
- 750 customers across 4 segments
- 170 products in 4 categories
- 8,000+ orders spanning 2021–2025
- 30 marketing campaigns, 43 geographic locations across 5 countries

Dataset Description

Data Sources

Table	Type	Rows	Key Fields
Sales_Fact	Fact	8,014	Order_ID, Customer_ID, Product_ID, Sales, Profit, Discount
Customer_Transactions	Fact	750	Customer_ID, Total_Orders, Total_Sales, Avg_Order_Value
Customer_Dimension	Dimension	810	Customer_ID, Customer_Name, Segment, Signup_Date
Product_Dimension	Dimension	170	Product_ID, Product_Name, Category, Sub_Category, Price
Geography_Dimension	Dimension	43	Geography_ID, Country, Region, State, City
Time_Dimension	Dimension	1,826	Time_ID, Date, Month, Quarter, Year
Marketing_Campaign	Dimension	30	Campaign_ID, Campaign_Name, Campaign_Type

Table 1: Data source summary (7 tables in star schema)

Data Quality Issues

Issue	Table	Count	Resolution
Duplicate rows	Sales_Fact	146	Removed (kept first)
Missing Discount	Sales_Fact	163	Filled with 0
Missing Ship_Mode	Sales_Fact	163	Filled with "Standard Class"
Missing Segment	Customer_Dim	22	Filled with "Unknown"
Missing Sub_Category	Product_Dim	3	Filled with "Uncategorized"
Missing Email	Customer_Dim	22	Filled with placeholder

Table 2: Data quality issues identified and resolved

Star Schema Architecture

Schema Design

The dataset follows a **Star Schema** with `Sales_Fact` at the center, connected to five dimension tables via foreign keys using Tableau's Logical Layer (Relationships).

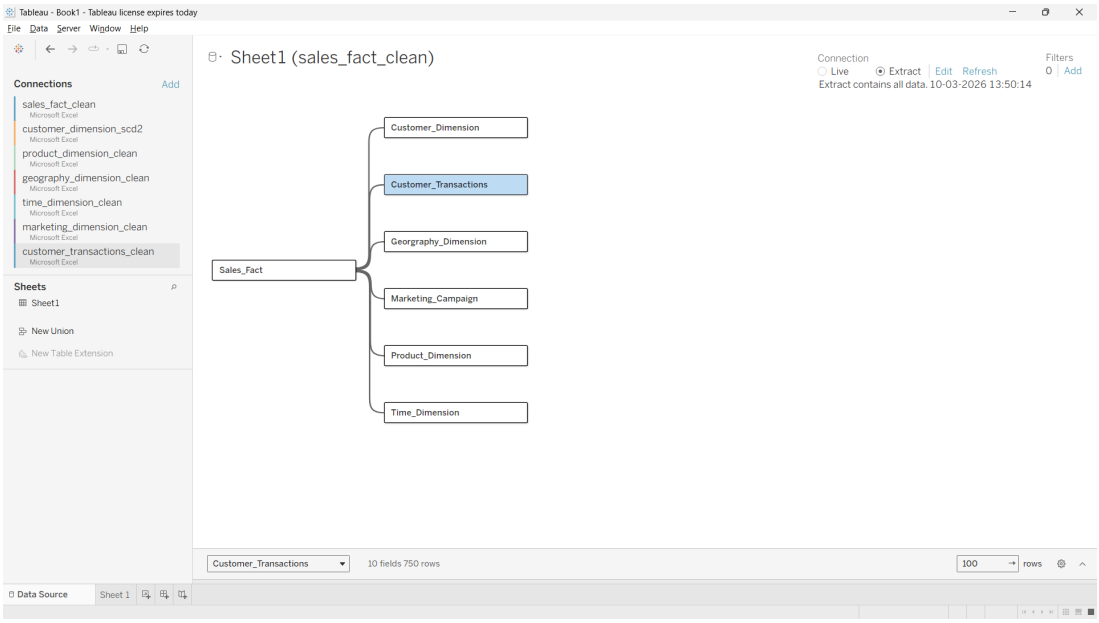


Figure 1: Star Schema Data Model in Tableau

Logical Layer Relationships

Fact Field	Dimension Table	Dimension Field	Type
Customer_ID	Customer_Dimension	Customer_ID	Many-to-One
Product_ID	Product_Dimension	Product_ID	Many-to-One
Geography_ID	Geography_Dimension	Geography_ID	Many-to-One
Time_ID	Time_Dimension	Time_ID	Many-to-One
Campaign_ID	Marketing_Campaign	Campaign_ID	Many-to-One

Table 3: Relationship mapping between fact and dimension tables

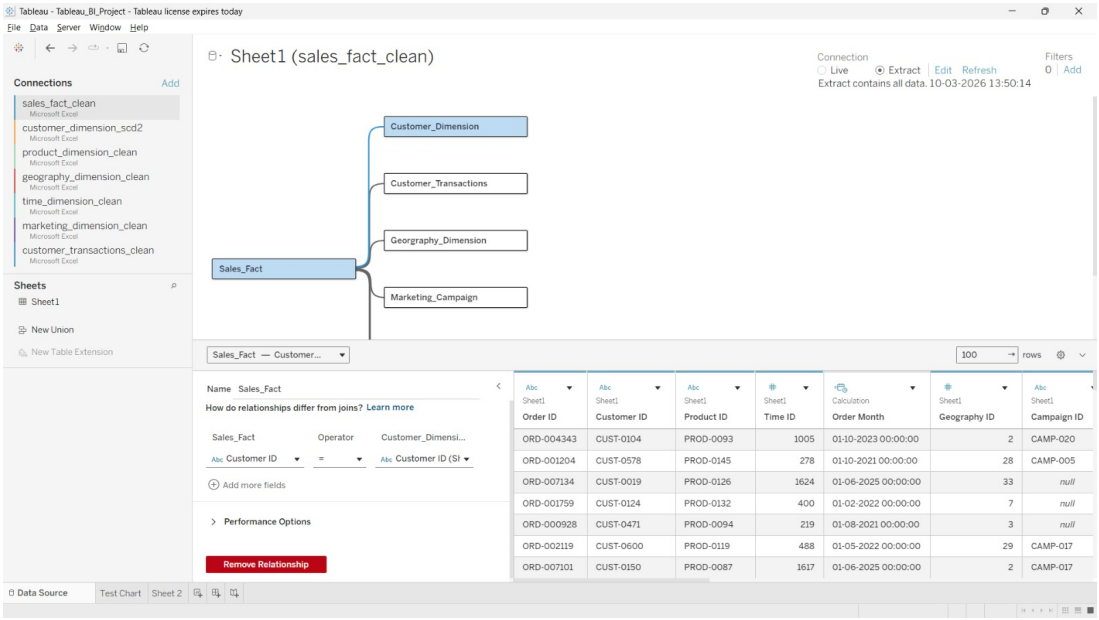


Figure 2: Relationship detail showing join clause

Why Relationships over Joins?

- Automatically handle join types per query context
- Avoid data duplication in memory
- Respect the grain of each table independently
- Handle null foreign keys gracefully (organic orders with no Campaign_ID)

Referential Integrity

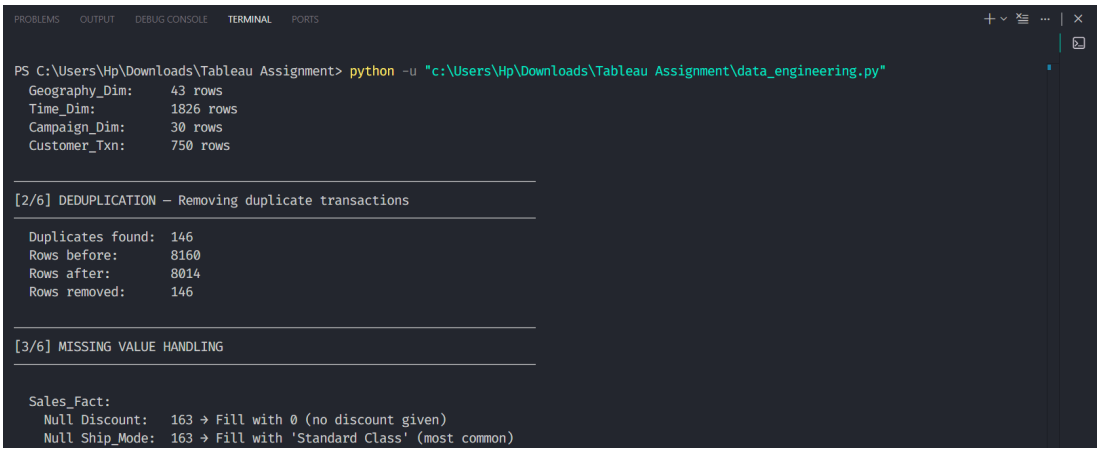
All foreign keys validated: 0 orphan Customer_IDs, 0 orphan Product_IDs. Campaign_ID nulls represent organic (non-campaign) orders, labeled as "Organic / No Campaign" via Tableau aliases.

Data Engineering

Deduplication

146 duplicate transaction rows were identified and removed using Python's `drop_duplicates()` method to ensure data integrity.

Before: 8,160 rows → **After:** 8,014 rows



```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS C:\Users\Hp\Downloads\Tableau Assignment> python -u "c:\Users\Hp\Downloads\Tableau Assignment\data_engineering.py"
  Geography_Dim:     43 rows
  Time_Dim:          1826 rows
  Campaign_Dim:      30 rows
  Customer_Txn:      750 rows

──────────────────────────────────────────────────────────────────────
[2/6] DEDUPLICATION — Removing duplicate transactions
──────────────────────────────────────────────────────────────────────

  Duplicates found:  146
  Rows before:       8160
  Rows after:        8014
  Rows removed:      146

──────────────────────────────────────────────────────────────────────
[3/6] MISSING VALUE HANDLING
──────────────────────────────────────────────────────────────────────

  Sales_Fact:
    Null Discount:   163 → Fill with 0 (no discount given)
    Null Ship_Mode:  163 → Fill with 'Standard Class' (most common)
```

Figure 3: Data engineering pipeline output showing deduplication results

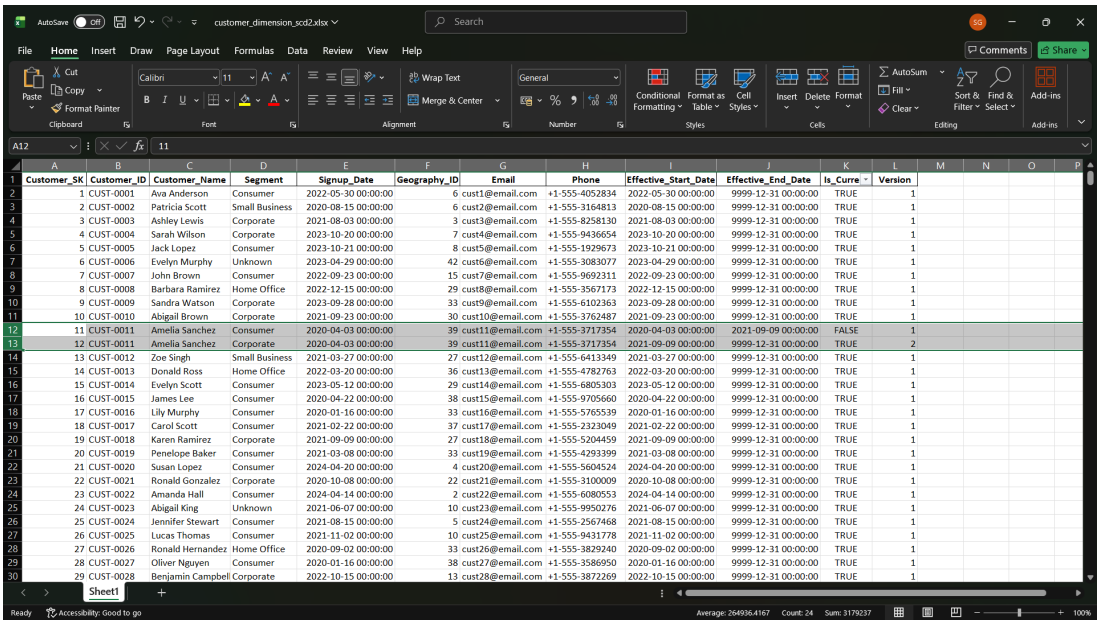
Slowly Changing Dimensions (SCD Type 2)

What is SCD Type 2?

SCD Type 2 preserves complete history of dimension attribute changes by creating **new rows with effective date ranges** instead of overwriting existing values. This enables historical analysis of how customer segments evolved over time.

Implementation details:

- 60 customers (∼8%) had segment changes over the data period
- Original rows closed (`Effective_End_Date` set, `Is_Current = False`)
- New rows created with updated segment and `Is_Current = True`
- Each row received a unique Surrogate Key (`Customer_SK`)

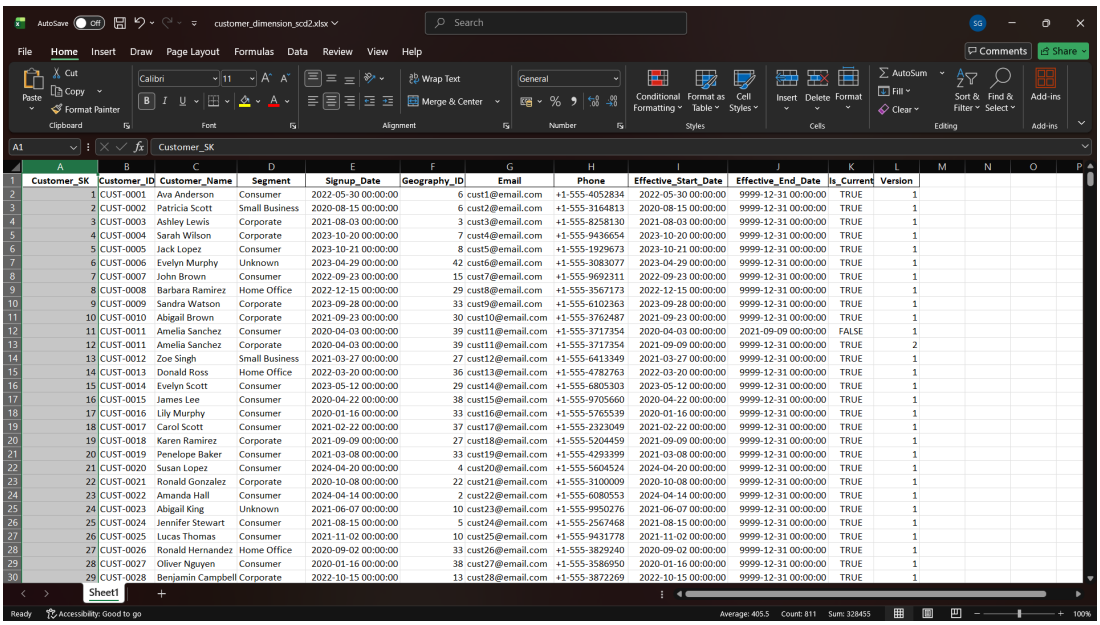


Customer_SK	Customer_ID	Customer_Name	Segment	Signup_Date	Geography_ID	Email	Phone	Effective_Start_Date	Effective_End_Date	Is_Curre	Version
1	CUST-0001	Ava Anderson	Consumer	2022-05-30 00:00:00	6	cust1@email.com	+1-555-4052834	2022-05-30 00:00:00	9999-12-31 00:00:00	TRUE	1
2	CUST-0002	Patricia Scott	Small Business	2020-08-15 00:00:00	6	cust2@email.com	+1-555-3164813	2020-08-15 00:00:00	9999-12-31 00:00:00	TRUE	1
3	CUST-0003	Ashley Lewis	Corporate	2021-08-03 00:00:00	3	cust3@email.com	+1-555-8258130	2021-08-03 00:00:00	9999-12-31 00:00:00	TRUE	1
4	CUST-0004	Sarah Wilson	Corporate	2023-10-20 00:00:00	7	cust4@email.com	+1-555-9436654	2023-10-20 00:00:00	9999-12-31 00:00:00	TRUE	1
5	CUST-0005	Jack Lopez	Consumer	2023-10-21 00:00:00	8	cust5@email.com	+1-555-1929673	2023-10-21 00:00:00	9999-12-31 00:00:00	TRUE	1
6	CUST-0006	Evelyn Murphy	Unknown	2023-04-29 00:00:00	42	cust6@email.com	+1-555-3083077	2023-04-29 00:00:00	9999-12-31 00:00:00	TRUE	1
7	CUST-0007	John Brown	Consumer	2022-09-23 00:00:00	15	cust7@email.com	+1-555-9692311	2022-09-23 00:00:00	9999-12-31 00:00:00	TRUE	1
8	CUST-0008	Barbara Ramirez	Home Office	2022-12-15 00:00:00	29	cust8@email.com	+1-555-3567173	2022-12-15 00:00:00	9999-12-31 00:00:00	TRUE	1
9	CUST-0009	Sandra Watson	Corporate	2023-09-28 00:00:00	33	cust9@email.com	+1-555-6102363	2023-09-28 00:00:00	9999-12-31 00:00:00	TRUE	1
10	CUST-0010	Abigail Brown	Corporate	2021-09-23 00:00:00	30	cust10@email.com	+1-555-3762487	2021-09-23 00:00:00	9999-12-31 00:00:00	TRUE	1
11	CUST-0011	Amelia Sanchez	Consumer	2020-04-03 00:00:00	39	cust11@email.com	+1-555-3717354	2020-04-03 00:00:00	2021-09-09 00:00:00	FALSE	1
12	CUST-0011	Amelia Sanchez	Corporate	2020-04-03 00:00:00	39	cust11@email.com	+1-555-3717354	2021-09-09 00:00:00	9999-12-31 00:00:00	TRUE	2
13	CUST-0012	Zoe Singh	Small Business	2021-03-27 00:00:00	27	cust12@email.com	+1-555-6413349	2021-03-27 00:00:00	9999-12-31 00:00:00	TRUE	1
14	CUST-0013	Donald Ross	Home Office	2022-03-20 00:00:00	36	cust13@email.com	+1-555-4782763	2022-03-20 00:00:00	9999-12-31 00:00:00	TRUE	1
15	CUST-0014	Evelyn Scott	Consumer	2023-05-12 00:00:00	29	cust14@email.com	+1-555-6805303	2023-05-12 00:00:00	9999-12-31 00:00:00	TRUE	1
16	CUST-0015	James Lee	Consumer	2020-04-22 00:00:00	38	cust15@email.com	+1-555-9705660	2020-04-22 00:00:00	9999-12-31 00:00:00	TRUE	1
17	CUST-0016	Lily Murphy	Consumer	2020-01-16 00:00:00	33	cust16@email.com	+1-555-5765539	2020-01-16 00:00:00	9999-12-31 00:00:00	TRUE	1
18	CUST-0017	Carol Scott	Consumer	2021-02-22 00:00:00	37	cust17@email.com	+1-555-2323049	2021-02-22 00:00:00	9999-12-31 00:00:00	TRUE	1
19	CUST-0018	Karen Ramirez	Corporate	2021-09-09 00:00:00	27	cust18@email.com	+1-555-5204459	2021-09-09 00:00:00	9999-12-31 00:00:00	TRUE	1
20	CUST-0019	Penelope Baker	Consumer	2021-03-08 00:00:00	33	cust19@email.com	+1-555-4293399	2021-03-08 00:00:00	9999-12-31 00:00:00	TRUE	1
21	CUST-0020	Susan Lopez	Consumer	2024-04-20 00:00:00	4	cust20@email.com	+1-555-5604524	2024-04-20 00:00:00	9999-12-31 00:00:00	TRUE	1
22	CUST-0021	Ronald Gonzalez	Corporate	2020-10-08 00:00:00	22	cust21@email.com	+1-555-3100009	2020-10-08 00:00:00	9999-12-31 00:00:00	TRUE	1
23	CUST-0022	Amanda Hall	Consumer	2024-04-14 00:00:00	2	cust22@email.com	+1-555-6080553	2024-04-14 00:00:00	9999-12-31 00:00:00	TRUE	1
24	CUST-0023	Abigail King	Unknown	2021-06-07 00:00:00	10	cust23@email.com	+1-555-9950276	2021-06-07 00:00:00	9999-12-31 00:00:00	TRUE	1
25	CUST-0024	Jennifer Stewart	Consumer	2021-08-15 00:00:00	5	cust24@email.com	+1-555-2567468	2021-08-15 00:00:00	9999-12-31 00:00:00	TRUE	1
26	CUST-0025	Lucas Thomas	Consumer	2021-11-02 00:00:00	10	cust25@email.com	+1-555-9431778	2021-11-02 00:00:00	9999-12-31 00:00:00	TRUE	1
27	CUST-0026	Ronald Hernandez	Home Office	2020-09-02 00:00:00	33	cust26@email.com	+1-555-3829240	2020-09-02 00:00:00	9999-12-31 00:00:00	TRUE	1
28	CUST-0027	Oliver Nguyen	Consumer	2020-01-16 00:00:00	38	cust27@email.com	+1-555-3586950	2020-01-16 00:00:00	9999-12-31 00:00:00	TRUE	1
29	CUST-0028	Benjamin Campbell	Corporate	2022-10-15 00:00:00	13	cust28@email.com	+1-555-3872269	2022-10-15 00:00:00	9999-12-31 00:00:00	TRUE	1

Figure 4: SCD Type 2 example — same customer with two version rows

Surrogate Keys

Integer surrogate keys added to all dimension tables for faster joins, SCD2 support, and source-system independence.



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22	CUST-0021	Ronald Gonzalez	Corporate	2020-10-08 00:00:00	22	cust21@email.com	+1-555-3100009	2020-10-08 00:00:00	9999-12-31 00:00:00	TRUE	1
23	CUST-0022	Amanda Hall	Consumer	2024-04-14 00:00:00	2	cust22@email.com	+1-555-6080553	2024-04-14 00:00:00	9999-12-31 00:00:00	TRUE	1
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28	CUST-0027	Oliver Nguyen	Consumer	2020-01-16 00:00:00	38	cust27@email.com	+1-555-3586950	2020-01-16 00:00:00	9999-12-31 00:00:00	TRUE	1
29	CUST-0028	Benjamin Campbell	Corporate	2022-10-15 00:00:00	13	cust28@email.com	+1-555-3872269	2022-10-15 00:00:00	9999-12-31 00:00:00	TRUE	1

Figure 5: Surrogate key columns visible in processed data

Live vs Extract Connections

Feature	Live	Extract
Data freshness	Real-time	Snapshot (refresh needed)
Speed	Slower for large data	Much faster
Offline access	Not possible	Yes
Best for	Small data, real-time	Large data, dashboards

Table 4: Live vs Extract comparison — Extract used in this project

Hyper Engine Compression

Tableau's Hyper Engine uses **columnar storage**, **dictionary encoding**, **run-length encoding**, and **bit packing** for compression. Result: 10MB Excel → ∼2–3MB `.hyper` extract with 10–100× faster queries.

LOD Expressions

LOD (Level of Detail) expressions calculate at a different granularity than the visualization.

FIXED — Customer Lifetime Value

FIXED LOD Expression

`{ FIXED [Customer ID] : SUM([Sales]) }`
Calculates total lifetime revenue per customer at a fixed granularity, **independent of view filters** (except Context and Data Source filters).

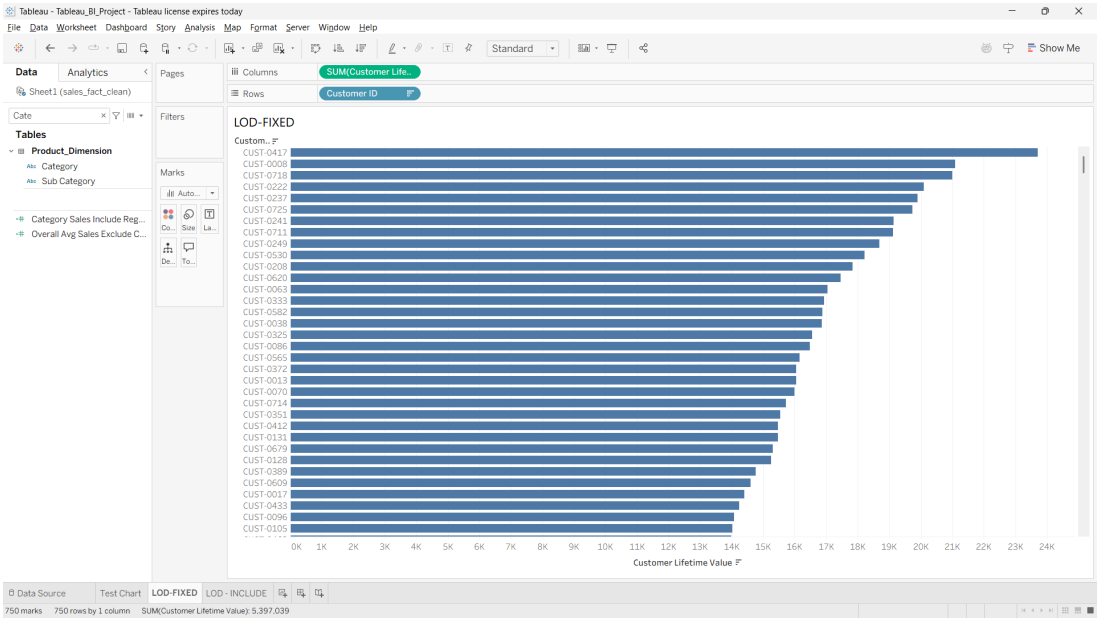

Figure 6: Customer Lifetime Value using FIXED LOD expression

INCLUDE

INCLUDE LOD Expression

`{ INCLUDE [Region] : SUM([Sales]) }`

Adds dimensions to the calculation beyond what is in the view, making it **more granular**.

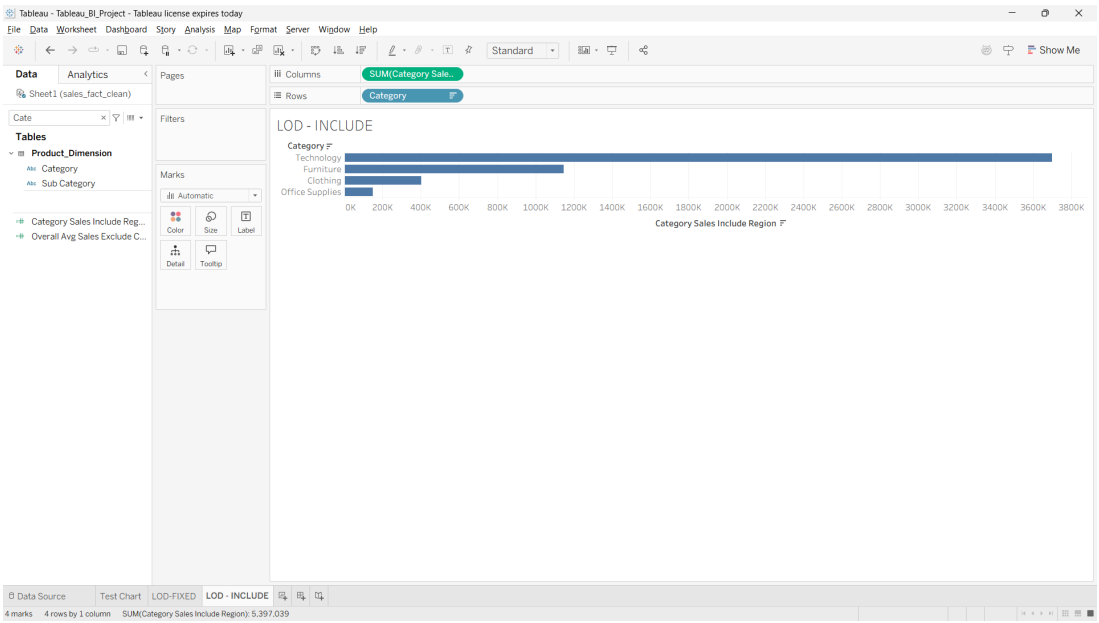

Figure 7: INCLUDE LOD — adds Region granularity to Category view

EXCLUDE

EXCLUDE LOD Expression

`{ EXCLUDE [Category] : AVG([Sales]) }`
Removes dimensions from the calculation, making it **less granular** — useful for comparing detail to overall averages.

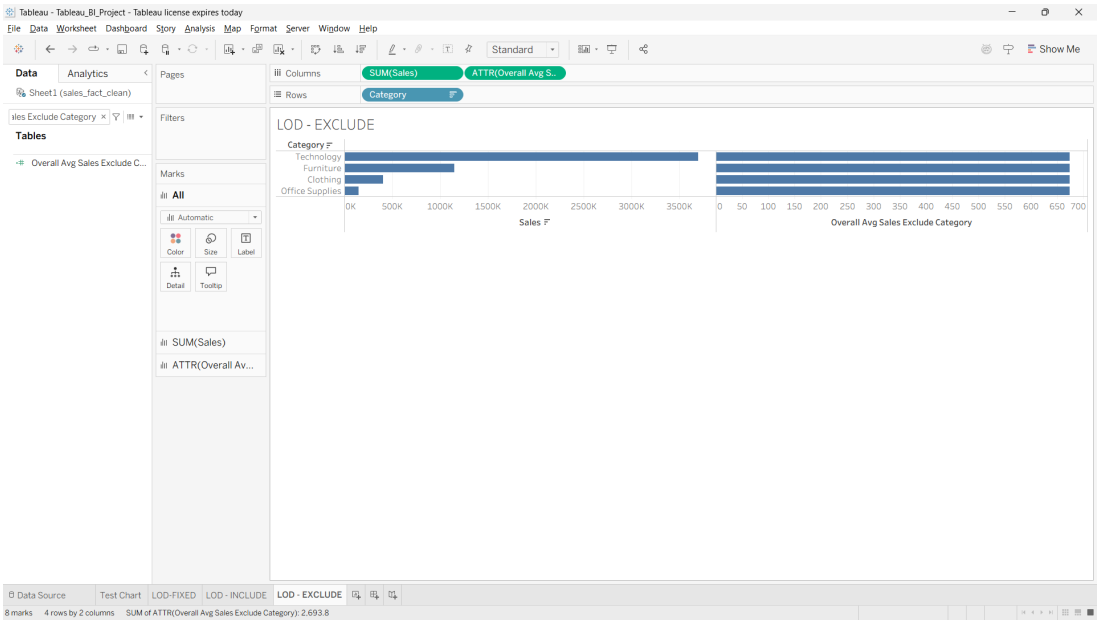

Figure 8: EXCLUDE LOD — overall average vs category-level sales

LOD Comparison & Performance

LOD Type	Granularity	Performance	Use Case
FIXED	Independent of view	Fastest	Customer-level metrics, KPIs
INCLUDE	Finer than view	Moderate	Adding hidden detail dimensions
EXCLUDE	Coarser than view	Slowest	Comparing to overall averages

Table 5: LOD expression comparison

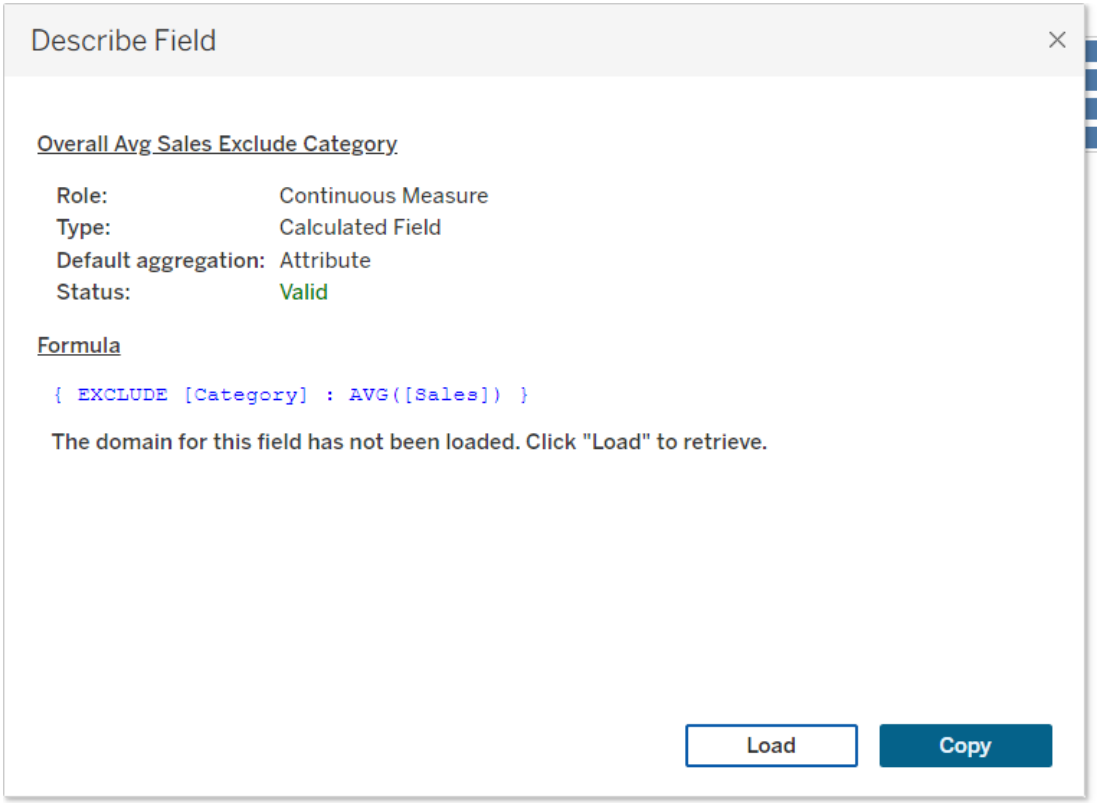

Figure 9: LOD calculation editor showing valid FIXED expression

Order of Operations

1. Extract Filters
2. Data Source Filters
3. **Context Filters**
4. **FIXED LOD calculations**
5. Dimension Filters
6. **INCLUDE / EXCLUDE LOD calculations**
7. Measure Filters
8. Table Calculations
9. Trend Lines, Reference Lines

FIXED LODs are **not** affected by dimension filters; INCLUDE/EXCLUDE are.

Cohort Analysis

Acquisition Month Cohort

Each customer is assigned to a cohort based on their first purchase month:

Acquisition Month (LOD)

```
DATETRUNC('month', { FIXED [Customer ID] : MIN([Order Date]) })
```

Retention Heatmap

Retention Rate

$$\text{Retention Rate} = \frac{\texttt{COUNTD([Customer ID])}}{\texttt{ATTR([Cohort Size])}}$$

$$\text{Where Cohort Size} = \{\ \texttt{FIXED [Acquisition Month] : COUNTD([Customer ID])}\ \}$$

The heatmap displays: Rows = Acquisition Month (cohort), Columns = Cohort Age (months since first purchase), Color = Retention percentage.

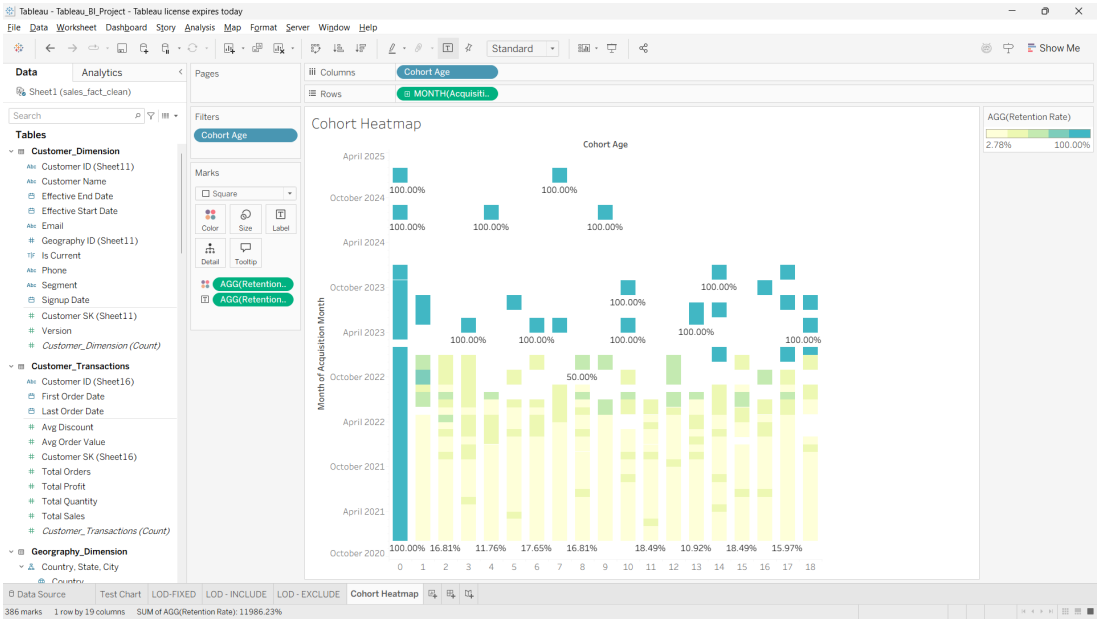

Figure 10: Cohort Retention Heatmap — darker cells indicate higher retention

Survival Curve

The survival curve shows cumulative retention over time — what percentage of each cohort remains active at each month.

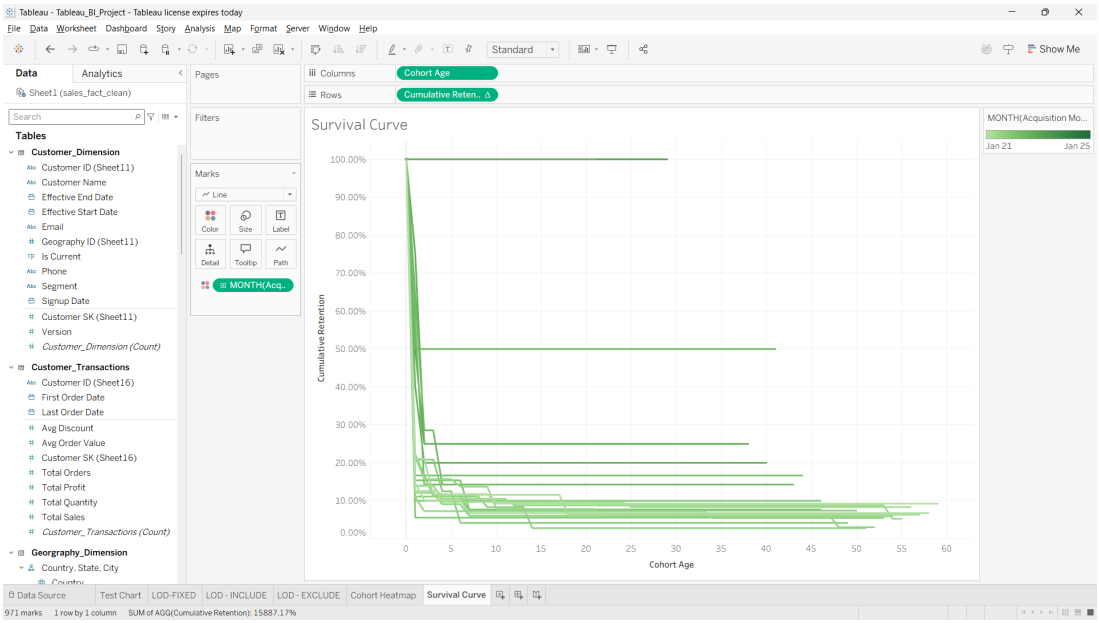

Figure 11: Survival Curve — each line represents one acquisition cohort

Business Insights

- All cohorts show rapid early drop-off (Month 0 to Month 5), then stabilize
- Older cohorts (2021–2022) demonstrate steadier long-term retention
- Q4 cohorts tend to show slightly higher retention, likely due to holiday promotions creating repeat buyers
- By Month 12, average retention across cohorts settles at approximately 10–15%

Predictive Forecasting

Forecast Setup

- **Method:** Exponential Smoothing (Tableau built-in)
- **Data:** Monthly aggregated sales
- **Forecast period:** 6–12 months ahead
- **Confidence band:** 95%

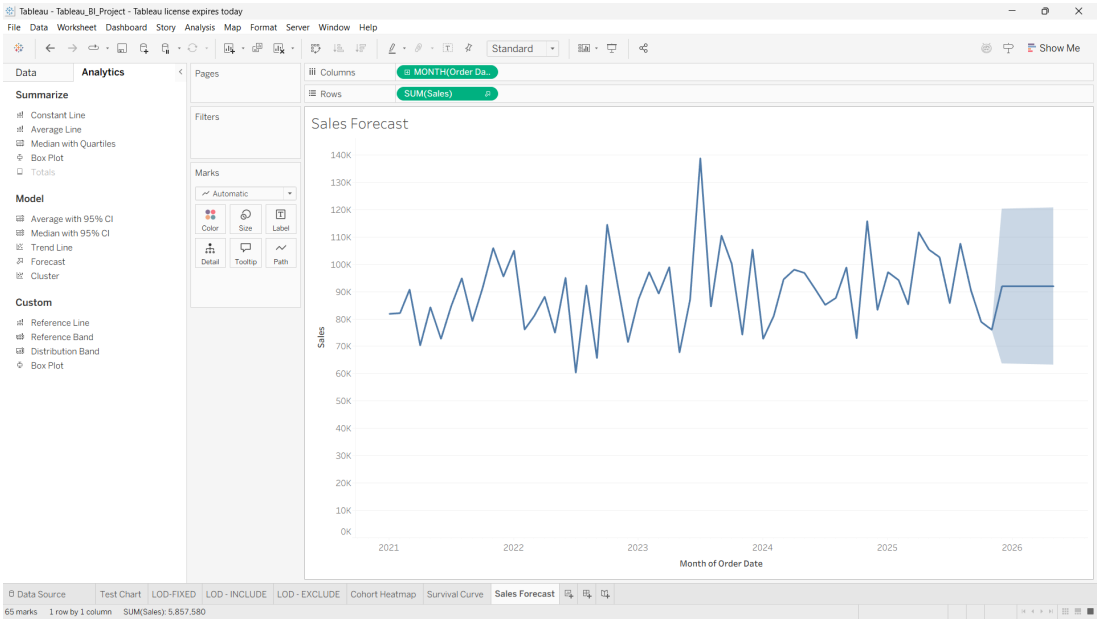

Figure 12: Sales Forecast with exponential smoothing and prediction intervals

Model Description

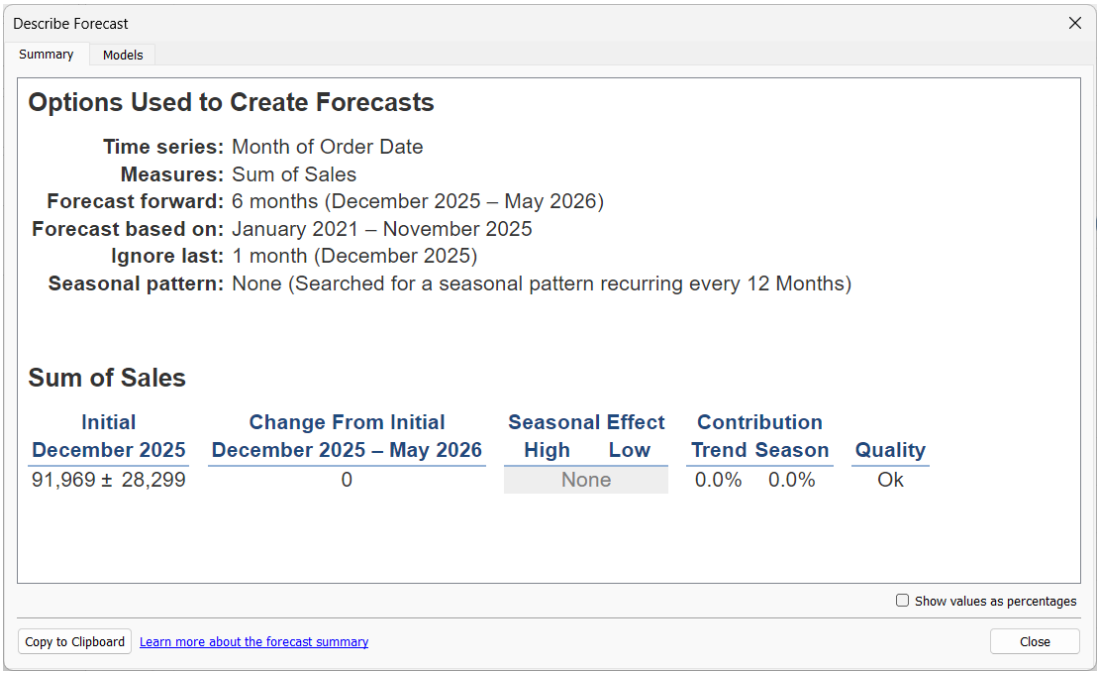

Figure 13: Forecast model details and quality metrics

MAPE — Forecast Accuracy

Mean Absolute Percentage Error

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{A_i - F_i}{A_i}\right| \times 100$$

$< 10\% =$ Excellent 10–20% = Good 20–50% = Reasonable $> 50\% =$ Inaccurate

Clustering & Machine Learning

K-Means Clustering

Tableau's built-in clustering groups customers using the K-Means algorithm based on:

- **Recency** — days since last purchase
- **Frequency** — number of orders
- **Monetary** — total spend

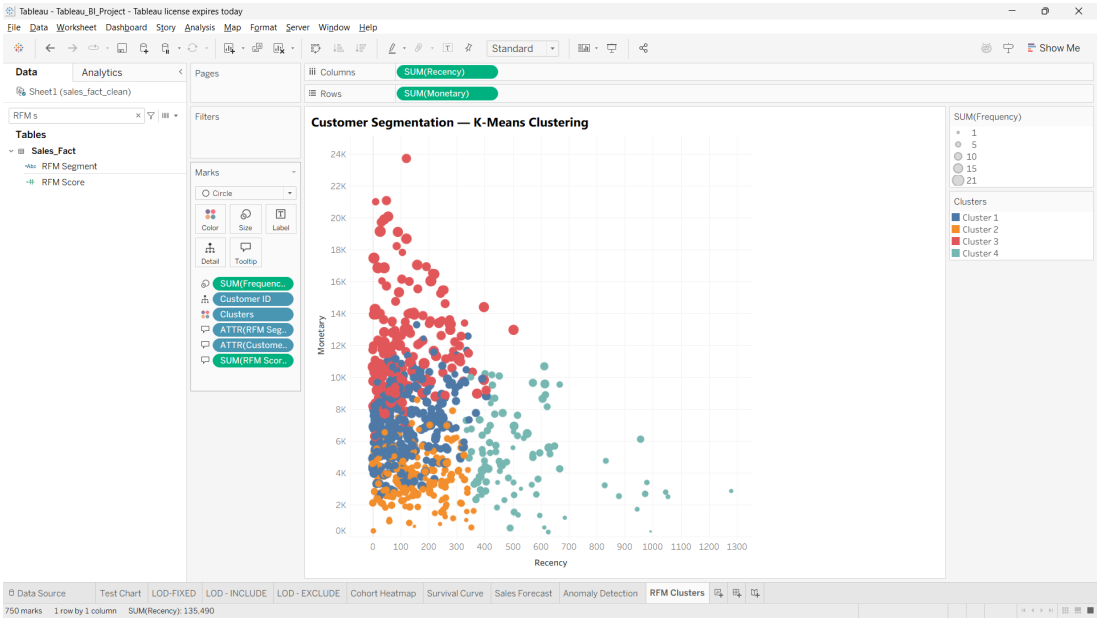

Figure 14: K-Means Cluster Scatter Plot (Recency vs Monetary, sized by Frequency)

RFM Segmentation

RFM Scoring

Component	Measure	Score
Recency	Days since last purchase	1 (old) to 5 (recent)
Frequency	Count of orders	1 (rare) to 5 (frequent)
Monetary	Total revenue	1 (low) to 5 (high)

Composite: RFM Score $= R \times 100 + F \times 10 + M$
e.g., 555 = Best customer, 111 = Worst

Segment	Criteria	Action
Champions	$R \geq 4,\ F \geq 4,\ M \geq 4$	Reward, upsell
Loyal Customers	$R \geq 4,\ F \geq 2$	Loyalty programs
At Risk	$R \leq 2,\ F \geq 3,\ M \geq 3$	Win-back campaigns
Hibernating	$R \leq 2,\ F \leq 2$	Re-engagement or release

Table 6: RFM segment labels with recommended business actions

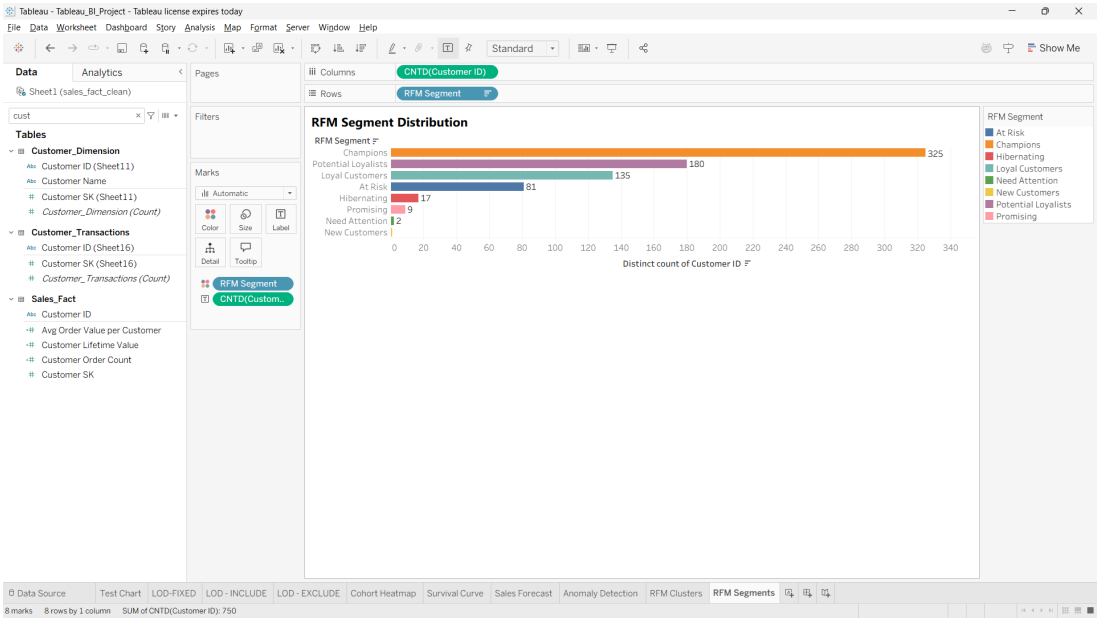

Figure 15: RFM Segment Distribution — count of customers per segment

RFM Index (weighted composite): $(R \times 0.35) + (F \times 0.35) + (M \times 0.30)$ — a single 1–5 score for overall customer ranking.

Statistical KPIs

Key statistical measures implemented as calculated fields in Tableau:

Z-Score Anomaly Detection

$$Z = \frac{X - \mu}{\sigma}$$

```
(SUM([Sales]) - WINDOW_AVG(SUM([Sales]))) / WINDOW_STDEV(SUM([Sales]))
```

$|Z| > 2$ flags anomalous months. Points outside 2 standard deviations warrant investigation.

Confidence Intervals (95%)

$$\text{Upper CI} = \bar{X} + 1.96 \times \frac{\sigma}{\sqrt{n}} \qquad \text{Lower CI} = \bar{X} - 1.96 \times \frac{\sigma}{\sqrt{n}}$$

Additional KPIs implemented:

- **3-Month Moving Average:** `WINDOW_AVG(SUM([Sales]), -2, 0)`
- **Rolling Std Dev:** `WINDOW_STDEV(SUM([Sales]), -5, 0)`
- **Profit Margin:** `SUM([Profit]) / SUM([Sales])`
- **AOV:** `SUM([Sales]) / COUNTD([Order ID])`

Dashboard Design

Three interactive dashboards were built, each targeting a different business audience.

Executive Control Panel

High-level KPI overview for management with interactive drill-down.

Components: KPI cards (Revenue, Profit, Orders, Customers, AOV), monthly sales trend, Top N products (parameter-driven), category treemap, regional performance.

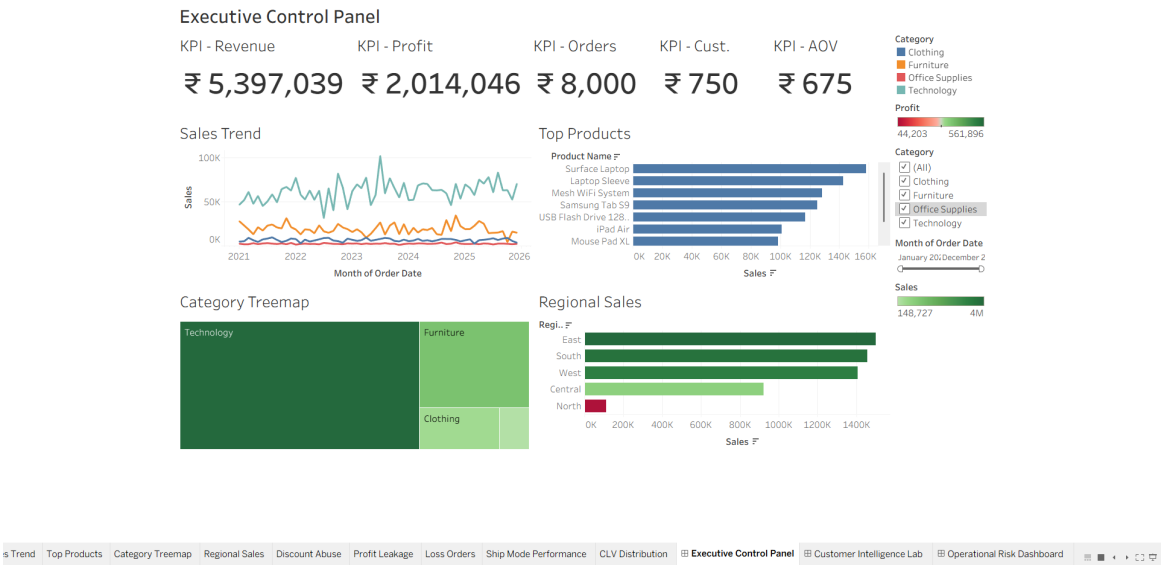

Figure 16: Executive Control Panel Dashboard

Key Insights:

- Technology category dominates revenue; Furniture shows higher profit margins
- Surface Laptop and Laptop Sleeve are the top revenue-generating products
- The Top N parameter enables dynamic exploration of product rankings

Customer Intelligence Lab

Deep customer analytics — cohort behavior, RFM segmentation, and lifetime value patterns.

Components: Cohort retention heatmap, RFM cluster scatter plot, RFM segment distribution, survival curve, CLV distribution.

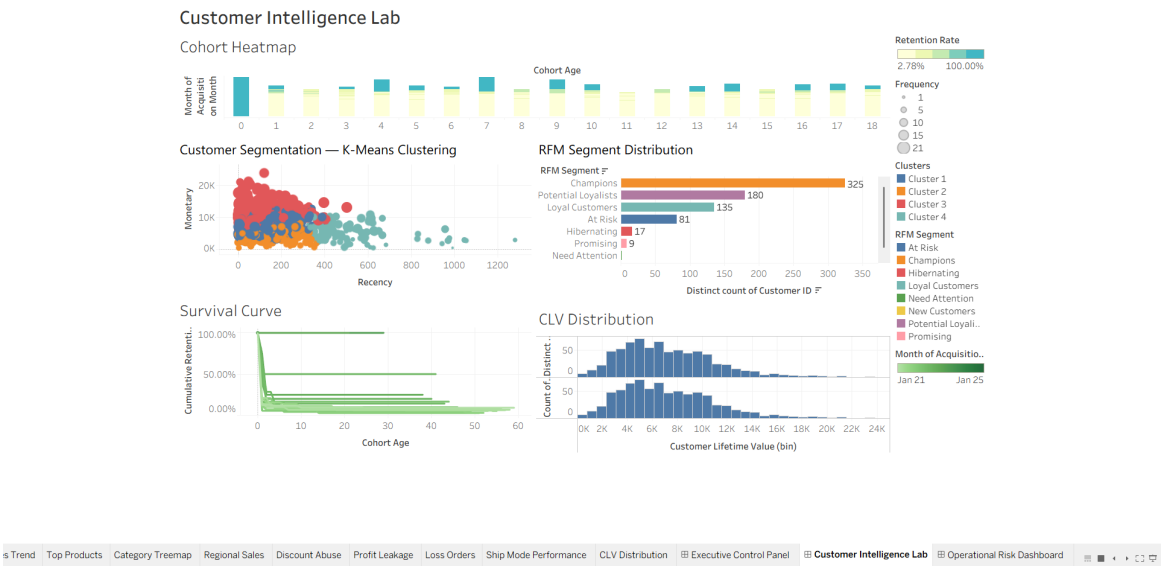

Figure 17: Customer Intelligence Lab Dashboard

Key Insights:

- Champions and Loyal Customers represent the highest-value segments

- Retention drops sharply in the first 3 months, then stabilizes
- K-Means clusters align well with RFM-based segmentation

Operational Risk Dashboard

Identifies anomalies, profit leakage, and discount abuse.

Components: Z-score anomaly detection, discount abuse scatter, profit leakage heatmap (Category × Region), loss orders table, shipping mode performance.

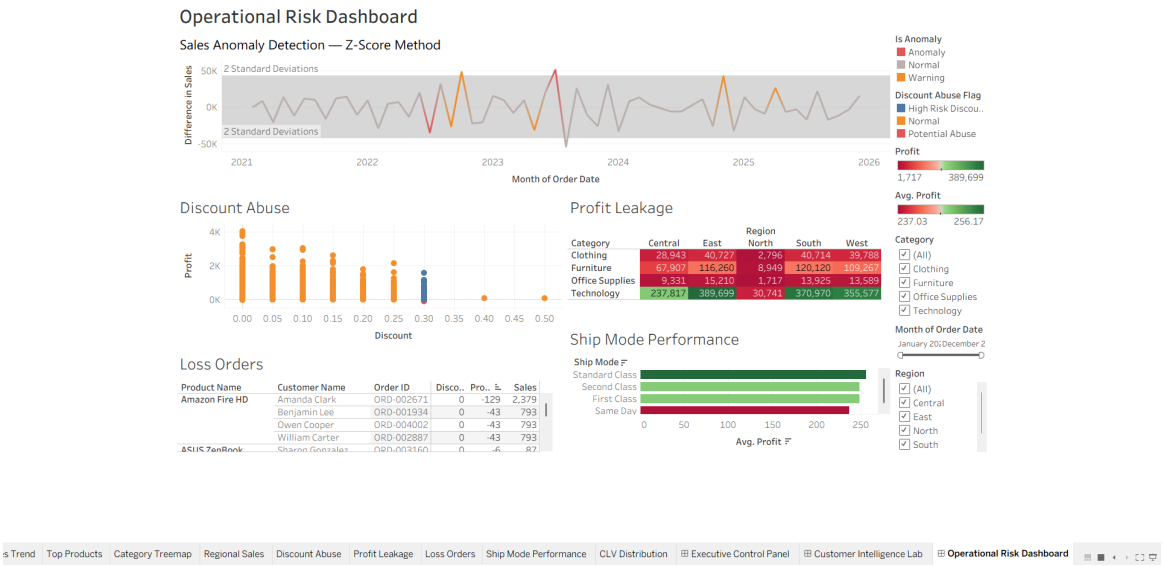

Figure 18: Operational Risk Dashboard

Key Insights:

- High discounts (>20%) correlate strongly with negative profits
- Specific category–region combinations show persistent profit leakage
- Anomalous sales months coincide with campaign periods

Performance Optimization

Performance Recording

Tableau's built-in Performance Recording was used to profile dashboard rendering:

1. Started recording via Help → Settings and Performance
2. Interacted with all three dashboards
3. Stopped recording and reviewed the performance workbook

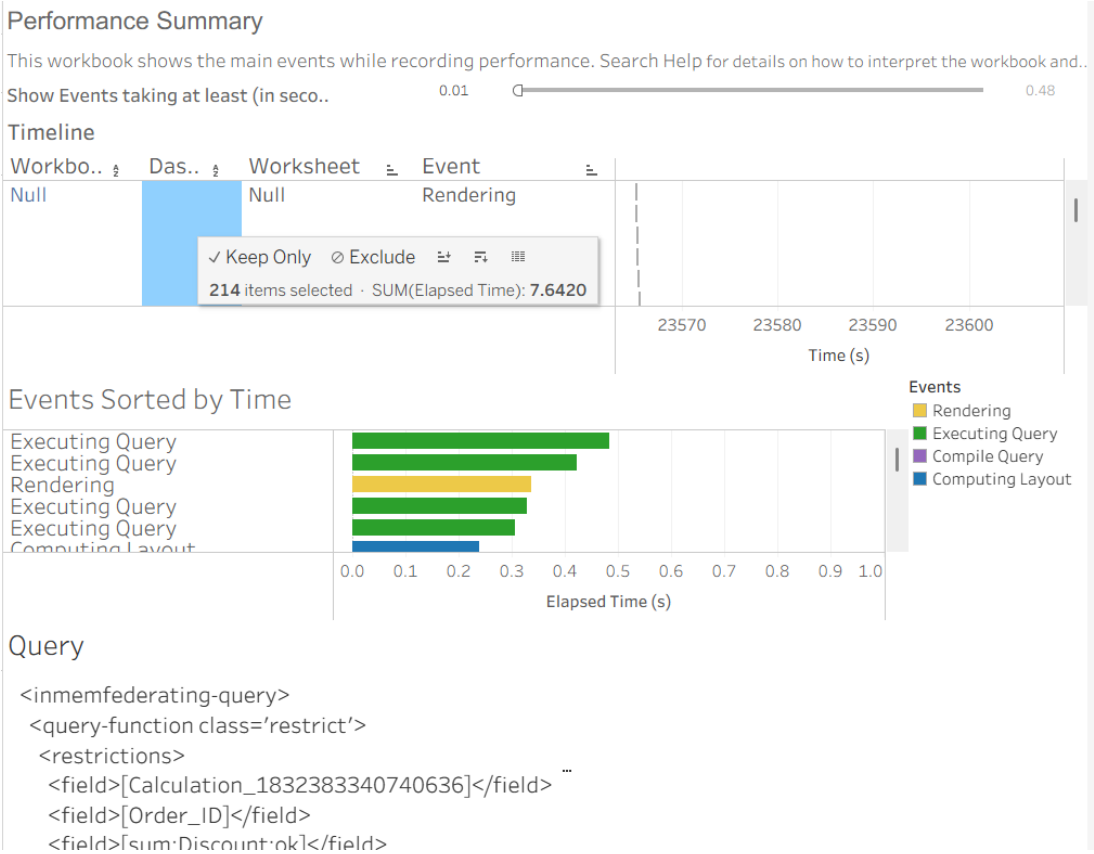

Figure 19: Performance Recording results — event timeline and elapsed times

Context Filters

Context filters execute **before** other filters, creating a temporary materialized table that speeds up subsequent filtering.

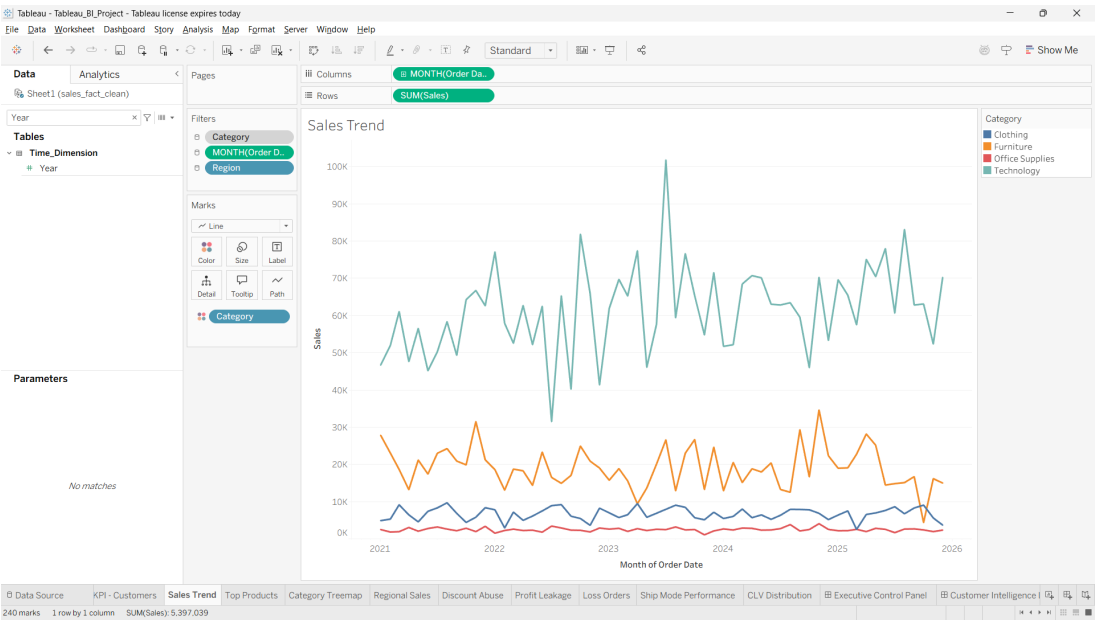

Figure 20: Context Filter (gray pill) applied to reduce data scope

Filter Hierarchy

Filter Type	When Applied	Speed Impact
Extract Filter	When extract is created	Fastest
Data Source Filter	At query time	Fast
Context Filter	Before other filters	Medium
Dimension Filter	Per visualization	Standard

Table 7: Filter types ordered by execution priority

Optimization Techniques Applied

- Extract connection instead of Live for faster querying
- Context filters on high-cardinality fields (Year)
- Surrogate keys (integer joins vs string joins)
- Discrete dates where possible to reduce cardinality
- Efficient LOD expressions (FIXED preferred for performance)

Theoretical Concepts

Tableau Architecture

Multi-Tier Architecture

1. Data Layer — Connects to sources (databases, files, cloud). Two modes: Live and Extract (.hyper). Hyper Engine handles compression and querying.

2. Analytics Layer (VizQL) — Proprietary Visual Query Language translates drag-and-drop into optimized queries. Handles calculations, LODs, and rendering pipeline.

3. Presentation Layer — Worksheets, Dashboards, and Stories provide the visual interface for end users.

Hyper Engine

Tableau's in-memory data engine (replaced TDE format):

- **Columnar storage** — column-by-column for faster analytical queries
- **Multi-threaded** — parallelizes queries across CPU cores
- **Compiled queries** — native machine code generation
- **Compression** — dictionary encoding, run-length encoding, bit packing
- Creates extracts 3× faster, queries 5–10× faster than TDE
- Supports billions of rows; file format: `.hyper`

Data Blending vs Joining

Feature	Joining	Data Blending
Scope	Within same data source	Across different sources
Granularity	Row-level detail preserved	Aggregated to common dimension
Performance	Faster (single query)	Slower (two queries)
When to use	Same source, need row detail	Different sources, different granularity

Table 8: Joining vs Data Blending comparison

Recommendation: Use Joins/Relationships for same-source data. Use Blending only for separate, incompatible sources.

LOD Expressions — Theory

LOD Expression Types

FIXED — Calculates at specified dimension level, ignores view dimensions. Only Context and Data Source filters apply. *Fastest performance.*

INCLUDE — Adds dimensions beyond the view, making calculation more granular. All filters apply.

EXCLUDE — Removes dimensions from the view, making calculation less granular. All filters apply. Useful for comparing to overall averages.

Tableau vs Power BI vs Qlik Sense

Feature	Tableau	Power BI	Qlik Sense
Vendor	Salesforce	Microsoft	Qlik
Strength	Best visualization	MS ecosystem integration	Associative data model
Language	VizQL, Calc Fields	DAX, M (Power Query)	Set Analysis, Script
Cost	$70/user/mo	$10–20/user/mo	$30/user/mo
Visualization	Superior, most customizable	Good, less flexible	Good, smart viz
ETL	Basic (Tableau Prep)	Built-in Power Query	Built-in data manager
AI/ML	Explain Data, clustering	AI Insights, AutoML	Insight Advisor
Best For	Complex analysis, dashboards	Budget BI, MS shops	Guided analytics

Table 9: BI tool comparison

Business Insights Summary

1. **Revenue Trend:** Sales show steady growth from 2021 to 2025 with seasonal peaks in Q4 driven by holiday demand.
2. **Top Products:** Surface Laptop and Laptop Sleeve dominate revenue, while accessories have higher margins.
3. **Customer Segments:** Corporate and Consumer segments contribute the most revenue; Home Office is smallest but growing.
4. **Geographic Performance:** The West region leads in total sales; certain Central sub-regions show profit leakage.
5. **Discount Impact:** Discounts exceeding 20% consistently result in negative profit margins.
6. **Cohort Retention:** Early cohorts (2021) show better long-term retention; all cohorts stabilize around 10–15% after 12 months.
7. **Anomalies:** Sales spikes align with major marketing campaigns; unexpected drops correlate with supply chain events.
8. **RFM Distribution:** ∼15% of customers are Champions, ∼25% Loyal, ∼20% At Risk, and ∼15% Hibernating.
9. **Forecast:** Exponential smoothing predicts continued moderate growth with seasonal Q4 peaks.
10. **Operational Risk:** High-discount orders in Office Supplies and specific regions are the primary profit leakage drivers.